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Method for manufacturing semiconductor device with deeply depleted channel

Abstract

A semiconductor device includes a substrate, a gate structure, a source region, a drain region, a doped region, and a channel region. The gate structure is disposed in the substrate, and the source region and drain regions being a first conductivity type respectively disposed at two sides of the gate structure. The doped region being a second conductivity type different from the first conductivity type is disposed below and separated from the gate structure, the source region, and drain region, the doped region. The channel region is disposed between the doped region and the gate structure and in contact with the doped region, and a dopant concentration of the channel region is less than a dopant concentration of the doped region.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5300447	12/1993	Anderson	N/A	N/A
5817558	12/1997	Wu	N/A	N/A
6093951	12/1999	Burr	N/A	N/A
6278165	12/2000	Oowaki	N/A	N/A
6452232	12/2001	Adan	N/A	N/A
2001/0030350	12/2000	Oowaki	N/A	N/A
2002/0014633	12/2001	Ashley	N/A	N/A
2002/0117725	12/2001	Oowaki	N/A	N/A
2003/0227061	12/2002	Yokogawa	257/E29.055	H10D 30/635
2006/0157797	12/2005	Tateshita	N/A	N/A
2009/0261350	12/2008	Yamamoto	257/77	H01L 29/7397
2010/0127329	12/2009	Park	N/A	N/A
2011/0180883	12/2010	Chen	257/E21.409	H01L 29/66575
2013/0126819	12/2012	Satoh	N/A	N/A
2013/0256791	12/2012	Iguchi	N/A	N/A
2015/0179791	12/2014	Kudou	257/77	H01L 21/31144
2016/0079417	12/2015	Iguchi	N/A	N/A
2016/0284696	12/2015	Ema	N/A	N/A

2020/0144055	12/2019	Yoshida	N/A	N/A
2021/0167191	12/2020	Gu	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
63-114174	12/1987	JP	N/A
63-114174	12/1987	JP	N/A
1-204472	12/1988	JP	N/A
1-204472	12/1988	JP	N/A
2-112284	12/1989	JP	N/A
2-112284	12/1989	JP	N/A
4-68540	12/1991	JP	N/A
4-68540	12/1991	JP	N/A
6-326308	12/1993	JP	N/A
2000-174283	12/1999	JP	N/A
2001-250950	12/2000	JP	N/A
2003-179223	12/2002	JP	N/A
2006-190821	12/2005	JP	N/A
2006-253706	12/2005	JP	N/A
2007-158321	12/2006	JP	N/A
2009-194392	12/2008	JP	N/A
2010-177318	12/2009	JP	N/A
2013-206945	12/2012	JP	N/A
2016-184655	12/2015	JP	N/A
2020-77712	12/2019	JP	N/A
9928975	12/1998	WO	N/A
2017/187720	12/2016	WO	N/A

OTHER PUBLICATIONS

Office action mailed on Jul. 22, 2022 for the U.S. Appl. No. 17/379,962, filed Jul. 19, 2021, pp. 1-7 ,Jul. 22, 2022. cited by applicant

Office action mailed on Sep. 1, 2022 for the U.S. Appl. No. 17/379,962, filed Jul. 19, 2021, pp. 1-17 ,Sep. 1, 2022. cited by applicant

Office action mailed on Feb. 15, 2023 for the U.S. Appl. No. 17/379,962, filed Jul. 19, 2021, pp. 1-12 ,Feb. 15, 2023. cited by applicant

Notice of Allowance mailed/issued on Jun. 28, 2023 for U.S. Appl. No. 17/379,962, filed Jul. 19, 2021, pp. 1-8. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a division of U.S. application Ser. No. 17/379,962, filed on Jul. 19, 2021. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure generally relates to a semiconductor device, and more particularly to a semiconductor device with a deeply depleted channel.

2. Description of the Prior Art

(2) Generally, a semiconductor device with deeply depleted channel is a device manufactured on a bulk silicon substrate. In the operation of such device, a depleted region can be generated below the gate structure at a predefined distance, which exhibits the electrical characteristic similar to that exhibited by a buried oxide of the silicon-on-insulator (SOI) substrate. Since the bulk silicon substrate is less expensive than SOI substrate, the semiconductor device with deeply depleted channel is becoming more and more popular in certain applications, such as IoT device applications with necessary of low power consumption.

(3) The semiconductor device with deeply depleted channel usually includes a planar gate structure and a highly doped region (also called screen region, SCR region). The SCR region is disposed under the planar gate structure at a predefined distance. However, there is often a current leakage generated between each source/drain region and the SCR region, which negatively affects the electrical performance of the semiconductor device.

SUMMARY OF THE INVENTION

(4) In view of this, there is still a need to provide an improved semiconductor device with deeply depleted channel and a manufacturing method thereof.

(5) According to some embodiments of the present disclosure, a semiconductor device is provided and includes a substrate, a gate structure, a source region, a drain region, a doped region, and a channel region. The gate structure is disposed in the substrate, and the source region and drain regions being a first conductivity type respectively disposed at two sides of the gate structure. The doped region being a second conductivity type different from the first conductivity type is disposed below and separated from the gate structure, the source region, and drain region. The channel region is disposed between the doped region and the gate structure and in contact with the doped region, and a dopant concentration of the channel region is less than a dopant concentration of the doped region.

(6) According to some embodiments of the present disclosure, a method for manufacturing a semiconductor device is provided and includes the following. A substrate is provided and a doped region is formed on the substrate. Then, a channel layer is formed on the doped region, and a trench is formed in the channel layer. Subsequently, a gate structure is formed in the trench, where the gate is disposed above and separated from the doped region. A source region and a drain region are then formed at two sides of the gate structure, where the source region and the drain region are disposed above and separated from the doped region. A dopant concentration of the channel layer is less than a dopant concentration of the doped region.

(7) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a schematic cross-sectional view illustrating a semiconductor device according to some embodiments of the present disclosure.

(3) FIG. 2 is a schematic enlarged cross-sectional view illustrating a concentration profile in a

region A of FIG. 1 according to some embodiments of the disclosure.

(4) FIG. 3 is a schematic cross-sectional view illustrating a semiconductor device according to some alternative embodiments of the present disclosure.

(5) FIG. 4 is a schematic cross-sectional view illustrating a semiconductor device according to some alternative embodiments of the present disclosure.

(6) FIG. 5 is a schematic top view illustrating a semiconductor device according to some embodiments of the present disclosure.

(7) FIGS. 6-9 are schematic cross-sectional views illustrating processes of a method for manufacturing a semiconductor device according to some embodiments of the disclosure.

(8) FIG. 10 is schematic cross-sectional view illustrating a process of a method for manufacturing a semiconductor device according to some alternative embodiments of the present disclosure.

DETAILED DESCRIPTION

(9) The present technology is described with reference to the attached figures, wherein like reference numerals are used throughout the figures to designate similar or equivalent elements. The figures are not drawn to scale and they are provided merely to illustrate the instant technology. Several aspects of the technology are described below with reference to example applications for illustration. It should be understood that numerous specific details, relationships, and methods are set forth to provide a full understanding of the technology. One having ordinary skill in the relevant art, however, will readily recognize that the technology can be practiced without one or more of the specific details or with other methods. In other instances, well-known structures or operations are not shown in detail to avoid obscuring the technology. The present technology is not limited by the illustrated ordering of acts or events, as some acts may occur in different orders and/or concurrently with other acts or events. Furthermore, not all illustrated acts or events are required to implement a methodology in accordance with the present technology.

(10) FIG. 1 is a schematic cross-sectional view illustrating a semiconductor device **100-1** according to some embodiments of the present disclosure. Referring to FIG. 1, a semiconductor device **100-1** may be a transistor with deeply depleted channel, and the semiconductor device **100-1** may include at least a substrate **102**, a doped region **104**, a channel region **110**, a gate structure **116**, a source region **122**, and a drain region **124**.

(11) The substrate **102** may be a semiconductor substrate of a required conductivity type, such as n-type or p-type. In the present disclosure, the substrate **102** may also be any type of substrate used for forming semiconductor devices thereon, including single crystal substrates, semiconductor on insulator (SOI) substrates, and epitaxial film on semiconductor (EPI) substrates, to name a few. Further, although the various embodiments will be primarily described with respect to materials and processes compatible with silicon-based semiconductor materials (e.g., silicon and alloys of silicon with germanium and/or carbon), the present technology is not limited in this regard. Rather, the various embodiments can be implemented using any types of semiconductor materials.

(12) The doped region **104** may be disposed on the substrate **102**, and the doped region **104** may be disposed below and separated from the gate structure **116**. The doped region **104** may have a concentration profile in a way that its dopant concentration gradually increases from the bottom of the doped region **104** to a predetermined region adjacent to the top of the doped region **104**, and then the dopant concentration gradually decreases from the predetermined region to the top of the doped region **104**. For the doped region **104** having a concentration profile, the peak concentration of the concentration profile may be 5×10^{18} to 1×10^{20} atoms/cm³, such as be equal to 5×10^{18} atoms/cm³, more than 5×10^{18} atoms/cm³ and less than 1×10^{20} atoms/cm³, or equal to 1×10^{20} atoms/cm³. The peak concentration of the doped region **104** may extend laterally in a horizontal region below the source region **122**, the drain region **124**, and the gate structure **116**. For an n-type semiconductor device, such as silicon-based NMOS, the doped region **104** may include p-type dopants, such as boron (B) or indium (I). In contrast, for a p-type semiconductor device, such as silicon-based PMOS, the doped region **104**

may include n-type dopants, such as arsenic (As), antimony (Sb) or phosphorous (P). It is noted that, for a case where the doped region **104** is formed by performing an implantation process on the substrate **102**, the doped region **104** may be regarded a doped region disposed in an upper portion of the substrate **102**. In contrast, for a case where the doped region **104** is formed by performing an epitaxial growth process on the substrate **102**, the doped region **104** may be regarded a doped region disposed on a top surface of the substrate **102**.

(13) The channel region **110** is disposed on the doped region **104** in a way that the source and drain regions **122**, **124** may be separated from the doped region **104** by the channel region **110**. Besides, portions of the channel region **110** may be disposed between the doped region **104** and the gate structure **116**. Therefore, the gate structure **116** is also separated from the doped region **104** by the channel region **110**. The channel region **110** may be an undoped or slightly doped region, such as undoped or slightly doped silicon, so that an average dopant concentration of the channel region **110** is less than an average dopant concentration of the doped region **104**.

(14) According to some embodiments of the present disclosure, the channel region **110** may be a single-layered structure or a double-layered structure. For a double-layered channel region **110**, the channel region **110** may include a first channel layer **106** and a second channel layer **108** in contact with the underlying doped region **104**. The first channel layer **106** may be an undoped or slightly doped layer laterally extending on the doped region **104**. A trench **140** may be formed in the first channel layer **106** and extend to an upper portion of the doped region. Besides, the second channel layer **108** may be an undoped or slightly doped layer which conforms to the surface of the trench **140**. Since the second channel layer **108** is disposed in the trench **140**, portions of the second channel layer **108** may be regarded as being embedded in or surrounded by the doped region **104**. According to different requirements, the trench **140** may be designed to include a curved surface or to include a surface including a flat bottom and slanted sidewalls. In either case, the second channel layer **108** may conform to the surface of the trench **140**. According to some embodiments of the present disclosure, the thickness of the first channel layer **106** may be greater than the thickness of the second channel layer **108**. For example, the first channel layer **106** may be approximately 100-150 nm in thickness, and the second channel layer **108** may be approximately 10-40 nm in thickness.

(15) It should be noted that, although FIG. **1** illustrates a curved interface between the first channel layer **106** and the second channel layer **108**, the interface may be hard to be detected or observed when both the first channel layer **106** and the second channel layer **108** are made of the same material, such as undoped silicon.

(16) The gate structure **116** may be a trench gate including a gate dielectric **112** and a gate electrode **114**, and the gate structure **116** may be disposed in a gate trench **150** defined by an inner surface of the second channel layer **108**. Because of the existence of the second channel layer **108**, the gate structure **116** may be separated from the first channel layer **106** and the underlying doped region **104**. According to different requirements, the gate electrode **116** may be a polysilicon gate or a metal gate, but not limited thereto.

(17) The source region **122** and the drain region **124** may be disposed at two sides of the gate structure **116** respectively. Specifically, the conductivity type of each of the source region **122** and the drain region **124** is different from the conductivity type of the doped region **104**. For example, when the source region **122** and the drain region **124** are n-type regions (or p-type regions), the doped region **104** is p-type region (or n-type region).

(18) Besides, in order to further reduce the contact resistance, conductive regions made of silicide or other conductive material may be formed at or on the upper portions of the gate electrode **114**, the source region **122**, and the drain region **124**, respectively. For example, silicide regions **132**, **134** may be disposed on and electrically coupled to the source region **122** and the drain region **124**, respectively. A silicide region **136** may be disposed on the gate structure **116**, such as disposed on and electrically coupled to the gate electrode **114** of the gate structure **116**. The silicide region **136**

on the gate structure **116** may be laterally (i.e. along x-direction) separated from the silicide regions **132**, **134** respectively disposed on the source region **122** and the drain region **124**. In this way, electric current would not be directly transmitted between two adjacent silicide regions, thereby prevents possible leakage current in the semiconductor device **100-1**.

(19) Furthermore, in order to prevent the current (such as leakage) from flowing between adjacent devices, an isolation structure **130**, such as shallow trench isolation (STI), may be used to surround the active region of the semiconductor device **100-1**.

(20) During the operation of the semiconductor device **100-1**, the gate electrode **114** may be positively or negatively biased so that the carrier (e.g. hole or electron) may be transmitted between the source region **122** and the drain region **124** along a bottom surface of the gate dielectric **112**. A depleted region may be induced by the voltage bias from the gate electrode **114** and generated near the interface of the doped region **104** and the channel region **110**. Besides, since both of the source region **122** and the drain region **124** are disposed above and separated from the doped region **104**, the source region **122** and the drain region **124** are not in direct contact with the underlying doped region **104**. Therefore, current leakage between the source/drain regions **122**, **124** and the underlying doped region **104** could be reduced or avoided effectively. Besides, since the carrier transmitted under the gate structure **116** mainly flows in the second channel layer **108**, the electrical performance of the semiconductor device **100-1** may be well-controlled by adjusting the thickness of the second channel layer **108** to a predetermined value.

(21) FIG. 2 is a schematic enlarged cross-sectional view illustrating a concentration profile in a region A of FIG. 1 according to some embodiments of the disclosure. Referring to FIG. 2, the doped region **104** may have gradually changed dopant concentration from the bottom of the doped region **104** to a predetermined depth of the doped region **104**. Besides, the dopant concentration of the doped region **104** may be kept substantially the same along a lateral direction at every depth level. As illustrating in figure (a) and figure (b) on the right-hand side of FIG. 2, the concentration profile of the dopants in the doped region **104** may have at least one peak. The peak may have a maximum concentration at a predetermined depth (as illustrated in figure (a)) or have a maximum concentration distributed in a range along a depth direction (as illustrated in figure (b)). According to some embodiments of the present disclosure, in order to control the concentration profile of the dopants in the doped region **104**, dopants may be implanted into the doped region **104** by performing one or more times of ion implantation processes with different implantation energy and doses. Preferably, the dopant concentration of the doped region **104** may substantially have a range as $\pm 30\%$ of an average along the depth direction as a flat concentration profile, including the maximum concentration. More preferably, the dopant concentration of the doped region **104** may have a range as $\pm 20\%$ of the average along the depth direction as the flat concentration profile, including the maximum concentration. Furthermore preferably, the dopant concentration of the doped region **104** may have a range as $\pm 10\%$ of the average along the depth direction as the flat concentration profile, including the maximum concentration. Substantially, the flat concentration profile can be obtained with an appropriate condition of alternative performing ion implantations in several times according to selected accuracy of such range. It is noted that, the dopant concentration of the doped region **104** at a depth lower than the lowest point of the second channel layer **108** may have the maximum dopant concentration. In other words, the bottom surface of the second channel layer **108** under the gate structure **116** may be shallower than the depth of the maximum dopant concentration.

(22) FIG. 3 is a schematic cross-sectional view illustrating a semiconductor device according to some alternative embodiments of the present disclosure. Referring to FIG. 3, the semiconductor device **100-2** illustrated in FIG. 3 is similar to the semiconductor device **100-1** illustrated in FIG. 1, the main difference is that the channel region **110** is a single-layered structure rather than a double-layered structure. Thus, the trench **150** formed in the channel region **110** may not be covered with a thin layer of semiconductor, such as undoped silicon. Besides, the trench **150** may be regarded as

the gate trench **150** used to accommodate the gate structure **116**. According to the semiconductor device **100-2** illustrated in FIG. 3, the doped region **104** is also separated from the gate structure **116**, the source region **122**, and the drain region **124**. Since both of the source region **122** and the drain region **124** are disposed above and separated from the doped region **104**, the source region **122** and the drain region **124** are not in direct contact with the underlying doped region **104**. Therefore, current leakage between the source/drain regions **122**, **124** and the underlying doped region **104** could be reduced or avoided effectively.

(23) FIG. 4 is a schematic cross-sectional view illustrating a semiconductor device according to some alternative embodiments of the present disclosure. Referring to FIG. 4, the semiconductor device **100-3** illustrated in FIG. 4 is similar to the semiconductor device **100-1** illustrated in FIG. 1, the main difference is that portions of the gate electrode **114** with a width such as 20-200 nm is protruded from the gate trench **150** so that the top surface of the gate electrode is higher than the top surfaces of the source region **122** and the drain region **134**. Besides, spacers **118** with a width such as 10-50 nm may be disposed at two sides of the gate structure **116**, respectively. Therefore, the gate structure **116** may be laterally separated from the source and drain regions **122**, **124** by the spacers **118**. The silicide region **136** disposed on the gate structure **116** may be laterally (i.e. along x-direction) and vertically (i.e. along z-direction) separated from the silicide regions **132**, **134** respectively disposed on the source region **122** and the drain region **124**. In this way, electric current would not be directly transmitted between two adjacent silicide regions, thereby further preventing possible leakage current in the semiconductor device **100-3** rather than semiconductor device **100-1** illustrated in FIG. 1. Besides, because of the existence of the spacers **118** respectively disposed on two sidewalls of the gate electrode **114**, the silicide regions **132**, **134** may be disposed farther away from the sidewalls of the gate electrode **114** respectively as compared with the semiconductor device **100-1** illustrated in FIG. 1 that have no spacers.

(24) FIG. 5 is a schematic top view illustrating a semiconductor device **100-4** according to some embodiments of the present disclosure. Referring to FIG. 5, cross-sectional views respectively illustrated in FIG. 1, FIG. 3, FIG. 4 may be regarded as being taken along a line A-A' of FIG. 5. The active region (not shown) including at least the source region **122** and the drain region **124** may be surrounded by the isolation structure **130**. The source region **122** and the drain region **124** may extend along a first direction, such as x-direction, and the gate structure **116** may extend along a second direction, such as y-direction. It is noted that, the direction along which the source region **122** and the drain region **124** extend is not limited to be perpendicular to the direction along which the gate structure **116** extends.

(25) In order to enable one of ordinary skill in the art to implement the present disclosure, a method of manufacturing a semiconductor device of the present disclosure is further described below.

(26) FIGS. 6-9 are schematic cross-sectional views illustrating processes of a method for manufacturing a semiconductor device according to some embodiments of the disclosure. Referring to FIG. 6, a substrate **102**, such as semiconductor substrate, is provided. A doped well (not shown) of a certain conductivity type may be formed in the substrate **102**. According to some embodiments of the present disclosure, the doped well may be formed by performing at least one ion implantation process. For example, for a p-type doped well, p-type dopant such as boron may be implanted into the substrate **102** with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 100~300 keV, but not limited thereto. Alternatively, for an n-type doped well, n-type dopant such as phosphorus may be implanted into the substrate **102** with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 200~400 keV, but not limited thereto. A doped region **104** may be formed on the substrate **102** by an ion implantation process or by an epitaxial growth process. The conductivity type of the dopants in the doped region **104** is the same as the conductivity type of the dopants in the doped well. According to some embodiments of the present disclosure, the doped region **104** may be p-type region including p-type dopants, such as boron or indium. According to some alternative embodiments of the present disclosure, the doped region **104** may be n-type

region including n-type dopants, such as arsenic (As), antimony (Sb) or phosphorous (P). In order to control the concentration profile of the dopants, various kinds of dopants may be implanted at different energy and dosage in one or more ion implantation processes. Besides, the doped region **104** may also include other dopants, such as carbon, germanium, or boron fluoride, which may be used to adjust the concentration profile of the p-type dopants or n-type dopants. According to some embodiments of the present disclosure, for a case where the doped region **104** is p-type, the p-type doped region **104** may be formed by performing several ion implantation processes in sequence, such as implanting germanium into the p-type doped well with a dose of $1 \times 10^{13} \sim 1 \times 10^{15} \text{ cm}^{-2}$ at energy of 10~50 keV, implanting carbon into the p-type doped well with a dose of $1 \times 10^{13} \sim 1 \times 10^{15} \text{ cm}^{-2}$ at energy of 2~10 keV, implanting boron into the p-type doped well with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 10~30 keV, and implanting boron fluoride into the p-type doped well with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 200~400 keV. According to some alternative embodiments of the present disclosure, for a case where the doped region **104** is n-type, the n-type doped region **104** may also be formed by performing several ion implantation processes in sequence, such as implanting antimony into the n-type doped well with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 50~200 keV, and implanting antimony into the n-type doped well with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 10~30 keV. After the processes for forming the doped region **104**, the dopants in the doped region **104** may be activated by performing a heat treatment at a suitable temperature, such as in a range of 500~700° C.

(27) According to some embodiments of the present disclosure, the conditions and/or times of the ion implantation for forming the doped region **104** may be adjusted so as to obtain the required concentration profile, such as the concentration profile shown in FIG. 2. In order to obtain the concentration profile shown in FIG. 2(b), dopants with certain conductivity may be implanted into the substrate **102** by performing several ion implantation processes rather than performing a single ion implantation process. For example, for the p-type doped region **104**, p-type dopants such as boron fluoride (BF₂) may be implanted into the substrate **102** by performing four ion implantation processes in sequence, such as a first implantation process with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 5~15 keV, a second implantation process with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 7~17 keV, a third implantation process with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 10~20 keV, and a fourth implantation process with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 15~25 keV, but not limited thereto. Similarly, for the n-type doped region **104**, n-type dopants such as antimony (Sb) may be implanted into the substrate **102** by performing four ion implantation processes in sequence, such as a first implantation process with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 10~30 keV, a second implantation process with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 15~35 keV, a third implantation process with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 20~40 keV, and a fourth implantation process with a dose of $1 \times 10^{12} \sim 1 \times 10^{14} \text{ cm}^{-2}$ at energy of 25~45 keV, but not limited thereto. In order to obtain an accurate range (i.e. smaller variation) of the dopant concentration along the depth direction as the flat concentration profile, alternative ion implantations may be performed several times for obtaining desired accurate concentration range, but not limited thereto. Also, if it is enough to obtain the accurate range of the dopant concentration along the depth direction as the flat concentration profile, fewer times of ion implantations can be adopted, but not limited thereto.

(28) Still referring to FIG. 6, a first channel layer **106** with a thickness of 300-500 nm may be formed on the doped region **104** by an epitaxial growth process. The first channel layer **106** may be an undoped or slightly doped layer, and the dopant concentration of the first channel layer **106** is at least 10 to $10^{0.3}$ times less than the dopant concentration of the doped region **104**.

(29) Referring to FIG. 7, an isolation structure **130** may be formed in a predetermined region, and then a trench **140** with a width (along x direction) of 50-150 nm and a depth of (along z direction)

250-550 nm may be formed in the first channel layer **106** and down to an upper portion of the doped region **104**. Thus, a portion of the doped region **104** may be exposed from the bottom surface of the trench **140**. It is noted that, according to some alternative embodiments of the present disclosure, the trench **140** may not reach the doped region **104** so that none of the doped region **104** may be exposed from the bottom surface of the trench **140**. Moreover, even if none of the doped region **104** is exposed from the bottom surface of the trench **140** after the main process for forming the trench **140**, additional post-treatment, such as a wet etching process, may be performed on the bottom surface of the trench **140** until the doped region **104** is exposed from the bottom surface of the trench **140**. Besides, during the processes of forming the isolation structure **130** and the trench **140**, the top surface of the first channel layer **106** may be covered with a thin protection layer, such as SiO₂, in order to protect the top surface of the first channel layer **106** from damages.

(30) Referring to FIG. **8**, a second channel layer **108** with a thickness of 10-40 nm may be formed on the surface of the trench **140** to define a gate trench **150**. However, in a case where the first channel layer **106** is remained at the bottom surface of the trench **140**, the second channel layer **108** may be omitted and thus not be formed in the trench **140** according to some alternative embodiments of the present disclosure, such as the structure illustrated FIG. **3**. The lower portion of the second channel layer **108** may be in indirect contact with the doped region **104**. Besides, the second channel layer **108** may be an undoped or slightly doped layer, and the dopant concentration of the second channel layer **108** is at least 10 to 10³ times less than the dopant concentration of the doped region **104**. Then, a gate dielectric layer **142** may be conformally formed on the exposed surface of the first channel layer **106** and the second channel layer **108**. According to some embodiments of the present disclosure, the gate dielectric layer **142** with a thickness of 1-3 nm may be formed by thermal oxidation at a processing temperature of 800-1000° C. According to some alternative embodiments of the present disclosure, the gate dielectric layer **142** made of silicon oxynitride, hafnium oxide and the like may be formed by deposition process. Subsequently, a gate electrode layer **144**, which can be polysilicon or metal, is formed on the gate dielectric layer **142** and fills up the gate trench **150**. In order to fill the trench **150** completely, the thickness of the gate electrode layer **144** may be in a range of 500-700 nm.

(31) Referring to FIG. **9**, the gate electrode layer **144** may be planarized to form a gate electrode **114** in the gate trench **150** with stopping on the gate dielectric layer **142**. Afterwards, a source region **122** and a drain region **124** may be formed at sides of the gate electrode **114** by implanting dopants into the first channel layer **106** and the second channel layer **108**. According to some embodiments of the present disclosure, for a case where the source region **122** and the drain region **124** are both n-type, the source region **122** and drain region **124** may also be formed by performing at least one ion implantation processes, such as implanting phosphorus with a dose of 1×10¹⁵~1×10¹⁷ cm⁻² at energy of 5~15 keV. According to some alternative embodiments of the present disclosure, for a case where the source region **122** and the drain region **124** are both p-type, the source region **122** and drain region **124** may be formed by performing several ion implantation processes in sequence, such as implanting boron with a dose of 1×10¹²~1×10¹⁴ cm⁻² at energy of 5~15 keV, implanting germanium with a dose of 1×10¹⁴~1×10¹⁶ cm⁻² at energy of 10~30 keV, and implanting boron with a dose of 1×10¹⁴~1×10¹⁶ cm⁻² at energy of 3~10 keV. After the processes for forming the source region **122** and the drain region **124**, the dopants in the source region **122** and the drain region **124** may be activated by performing a heat treatment such as spike annealing at a suitable temperature, such as in a range of 1000~1100° C. Afterwards, silicidation processes may be carried out to form metal silicide respectively in the source region **122**, the drain region **124**, and the gate electrode **114**. Other processes may be performed afterwards so as to obtain the structure as illustrated in FIG. **1**.

(32) FIG. **10** is schematic cross-sectional view illustrating a process of a method for manufacturing a semiconductor device according to some alternative embodiments of the present disclosure.

Referring to FIG. 10, the process illustrated in FIG. 10 is an alternative process following FIG. 8. According to some alternative embodiments of present disclosure, following the step of FIG. 8, in order to manufacture the gate structure 116 as illustrated in FIG. 4, the gate electrode layer 144 may be patterned by photolithography process to form the gate electrode 114 illustrated in FIG. 10. The two opposite protruded sidewalls 118 of the gate electrode 114 (not illustrated) may be further formed with a width (along x-direction) of 50-150 nm as illustrated in FIG. 4 after forming the gate electrode 114, the source region 122 and the drain region 124 (not illustrated) may be further formed as illustrated in FIG. 4 after forming the two opposite protruded sidewalls 118, and the silicide regions 132, 134, 136 (not illustrated) may be further formed as illustrated in FIG. 4 after forming the source region 122 and the drain region 124. Thus, both the patterned gate electrode 114 and two opposite sidewalls 118 may laterally (i.e. along x-direction) and vertically (i.e. along z-direction) can separate between the silicide region 136 disposed on the gate structure 116 and the silicide regions 132, 134 respective disposed on the source region 122 and the drain region 124 to further prevent possible leakage current in the semiconductor device 100-3 (not illustrated in FIG. 10) as illustrated in FIG. 4 rather than semiconductor device 100-1 as illustrated in FIG. 1, because of separating more with spacers 118. Besides, it should be noted that the thickness and width of the gate electrode 114 may be adjusted according to different requirement and should not be interpreted in a limited sense.

(33) Afterwards, the source region 122 and the drain region 124 may be formed at sides of the gate electrode 114 by implanting dopants into the first channel layer 106 and the second channel layer 108. Thus, the top surface of the gate electrode 114 may be higher than the top surfaces of the source region 122 and the drain region 124. Afterwards, silicidation processes may be carried out to form metal silicide respectively in the source region 122, the drain region 124, and the gate electrode 114. Other processes may be performed afterwards so as to obtain the structure as illustrated in FIG. 4. It should be noted that, referring to FIG. 10, although each sidewall of the protruded portion of the gate electrode 114 are substantially aligned with the outer edge of the second channel layer 108, the sidewalls of the protruded portion of the gate electrode 114 may be laterally misaligned with the outer edge of the second channel layer 108. According to some embodiments of the present disclosure, the sidewall of the protruded portion of the gate electrode 114 may extend outwardly beyond the outer edge of the second channel layer 108 so that the gate electrode 114 may cover portions of the first channel layer 106, which prevent the doped region (i.e. S/D regions) from being formed in the portions of the first channel layer 106 that are covered by the gate electrode 114. According to some alternative embodiments of the present disclosure, the sidewall of the protruded portion of the gate electrode 114 may be trimmed inwardly relative to the outer edge of the second channel layer 108 so that portions of the doped region (i.e. S/D regions) may be formed in the portions of the second channel layer 108 that are not covered by the gate electrode 114.

(34) While various embodiments of the present technology have been described above, it should be understood that they have been presented by way of example only, and not limitation. Numerous changes to the disclosed embodiments can be made in accordance with the disclosure herein without departing from the spirit or scope of the technology. Thus, the breadth and scope of the present technology should not be limited by any of the above described embodiments. Rather, the scope of the technology should be defined in accordance with the following claims and their equivalents.

(35) Although the technology has been illustrated and described with respect to one or more implementations, equivalent alterations and modifications will occur to others skilled in the art upon the reading and understanding of this specification and the annexed drawings. In addition, while a particular feature of the technology may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

(36) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the technology. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Furthermore, to the extent that the terms “including”, “includes”, “having”, “has”, “with”, or variants thereof are used in either the detailed description and/or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising.”

(37) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this technology belongs. Also, the terms “about”, “substantially”, and “approximately”, as used herein with respect to a stated value or a property, are intended to indicate being within 20% of the stated value or property, unless otherwise specified above. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(38) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A method for manufacturing a semiconductor device, comprising: providing a substrate; forming a doped region on the substrate; forming a channel layer on the doped region; forming a trench in the channel layer; forming a gate structure in the trench, the gate being disposed above and separated from the doped region; and forming a source region and a drain region at two sides of the gate structure, the source region and the drain region being disposed above and separated from the doped region, wherein a dopant concentration of the channel layer is less than a dopant concentration of the doped region.
2. The method of claim 1, wherein the doped region is exposed from the trench when forming the trench in the channel layer.
3. The method of claim 2, wherein the trench extends into the doped region.
4. The method of claim 2, further comprising forming a further channel layer in the trench before forming the gate structure in the trench.
5. The method of claim 4, wherein the further channel layer conforms to a surface of the trench.
6. The method of claim 1, wherein the doped region is disposed below and separated from the gate structure, the source region, and drain region.
7. The method of claim 1, wherein the source region and the drain region are separated from the gate structure.
8. The method of claim 1, further comprising: forming first silicide regions disposed on the source region and the drain region respectively; and forming a second silicide region disposed on the gate structure, wherein each of the first silicide regions is laterally separated from the second silicide region.
9. The method of claim 1, wherein the source region and the drain region are a first conductivity type, and the doped region is a second conductivity type different from the first conductivity type.
10. The method of claim 1, wherein the doped region includes the dopant concentration to be equal to 5×10^{18} atoms/cm³, more than 5×10^{18} atoms/cm³ and less than 1×10^{20} atoms/cm³, or equal to 1×10^{20} atoms/cm³ at a first depth.
11. The method of claim 1, wherein the doped region includes a maximum of a dopant concentration at a first depth along a depth direction, and a bottom surface of the channel layer under the gate structure is shallower than the first depth.

12. The method of claim 1, wherein the doped region including the dopant concentration to be equal to 5×10^{18} atoms/cm.³, more than 5×10^{18} atoms/cm.³ and less than 1×10^{20} atoms/cm.³, or equal to 1×10^{20} atoms/cm.³ is disposed under the source region and the drain region.
13. The method of claim 4, wherein the doped region includes a maximum of a dopant concentration at a first depth along a depth direction, and a bottom surface of the further channel layer under the gate structure is shallower than the first depth.
14. The method of claim 1, wherein the trench surrounded by the channel layer, and the gate structure is disposed in the trench.
15. The method of claim 1, wherein a bottom surface of the trench is lower than bottom surfaces of the source region and the drain region.
16. The method of claim 1, wherein the doped region is entirely disposed below the gate structure, the source region, and the drain region.
17. The method of claim 1, further comprising: forming an isolation structure in the substrate, wherein the doped region is reached to the isolation structure.
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